Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Technology

**Vision of the Department**

*To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.*

**Mission of the Department**

*To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.*

**Session 2025-2026**

| | |
|---|---|
| **Vision:** *To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration* | **Mission:** *To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.* |

**Program Educational Objectives of the program (PEO):** (broad statements that describe the professional and career accomplishments)

| | | | |
|---|---|---|---|
| PEO1 | **Preparation** | **P: Preparation** | **Pep-CL abbreviation pronounce as Pep-si-lL easy to recall** |
| PEO2 | **Core Competence** | **E: Environment (Learning Environment)** | |
| PEO3 | **Breadth** | **P: Professionalism** | |
| PEO4 | **Professionalism** | **C: Core Competence** | |
| PEO5 | **Learning Environment** | **L: Breadth (Learning in diverse areas)** | |

**Program Outcomes (PO):** (statements that describe what a student should be able to do and know by the end of a program)

**Keywords of POs:**
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

**PSO Keywords:** Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

**Integrity:** I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

**Sheikh Wasimuddin**
**Name and Signature of Student and Date**

Sheikh.
21/02/26

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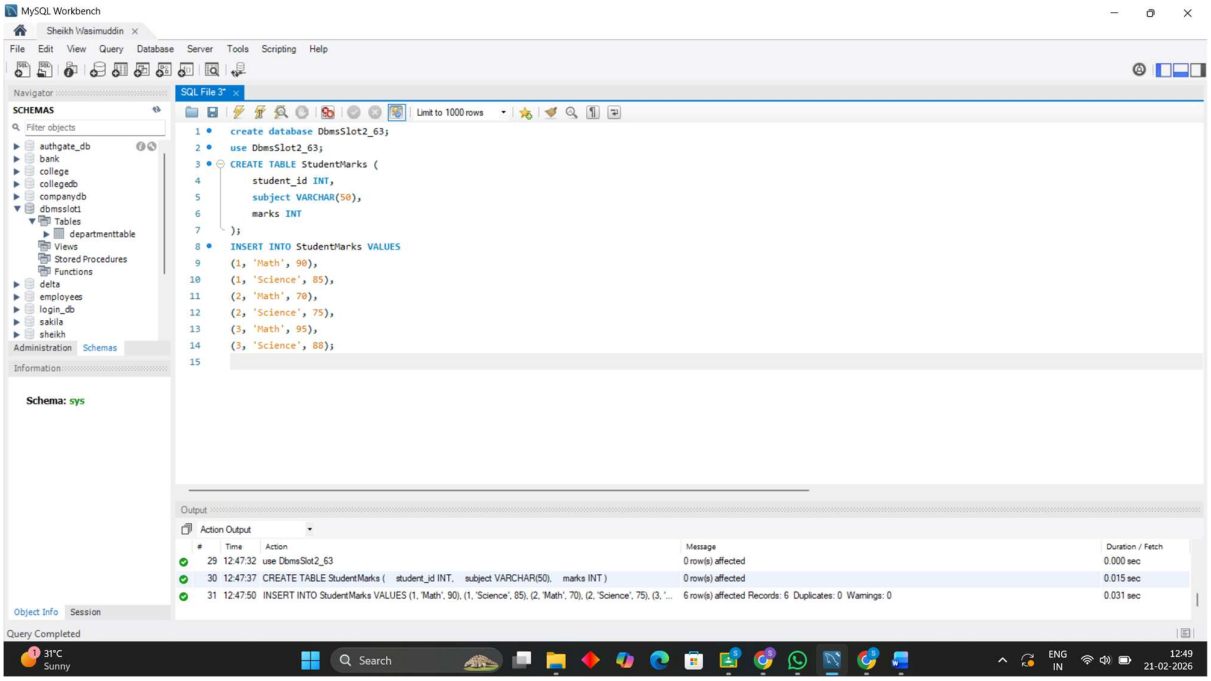
| Session | 2025-26 (EVEN) | Course Name | Competitive Coding Lab |
|---|---|---|---|
| Semester | 6 | Course Code | 23IOT1605 |
| Roll No | 63 | Name of Student | Sheikh Wasimuddin |
| Practical Number | DBMS SLOT-2 | | |

## Dataset02:StudentMarks

| student_id | subject | marks |
|---|---|---|
| 1 | Math | 90 |
| 1 | Science | 85 |
| 2 | Math | 70 |
| 2 | Science | 75 |
| 3 | Math | 95 |
| 3 | Science | 88 |



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## Practice questions:

### 1. Find students who scored above average in every subject.

```
SELECT student_id
FROM StudentMarks s
WHERE marks > (
    SELECT AVG(marks)
    FROM StudentMarks
    WHERE subject = s.subject
)
GROUP BY student_id
HAVING COUNT(*) = (SELECT COUNT(DISTINCT subject) FROM StudentMarks);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| student_id |
|---|
| 1 |
| 3 |

### 2. Find the topper in each subject.

```
SELECT student_id, subject, marks
FROM StudentMarks s
WHERE marks = (
    SELECT MAX(marks)
    FROM StudentMarks
    WHERE subject = s.subject
);
```

Result Grid | Filter Rows: | Export:

| student_id | subject | marks |
|---|---|---|
| 3 | Math | 95 |
| 3 | Science | 88 |

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## 3. Calculate overall percentage per student.

```
SELECT student_id,
       SUM(marks) AS total_marks,
       AVG(marks) AS percentage
FROM StudentMarks
GROUP BY student_id;
```

Result Grid | Filter Rows: | Export: | Wrap

| student_id | total_marks | percentage |
|---|---|---|
| 1 | 175 | 87.5000 |
| 2 | 145 | 72.5000 |
| 3 | 183 | 91.5000 |

## 4. Pivot the table to show subjects as columns.

```
SELECT student_id,
       MAX(CASE WHEN subject = 'Math' THEN marks END) AS Math,
       MAX(CASE WHEN subject = 'Science' THEN marks END) AS Science
FROM StudentMarks
GROUP BY student_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| student_id | Math | Science |
|---|---|---|
| 1 | 90 | 85 |
| 2 | 70 | 75 |
| 3 | 95 | 88 |

## 5. Find students who scored same marks in all subjects.

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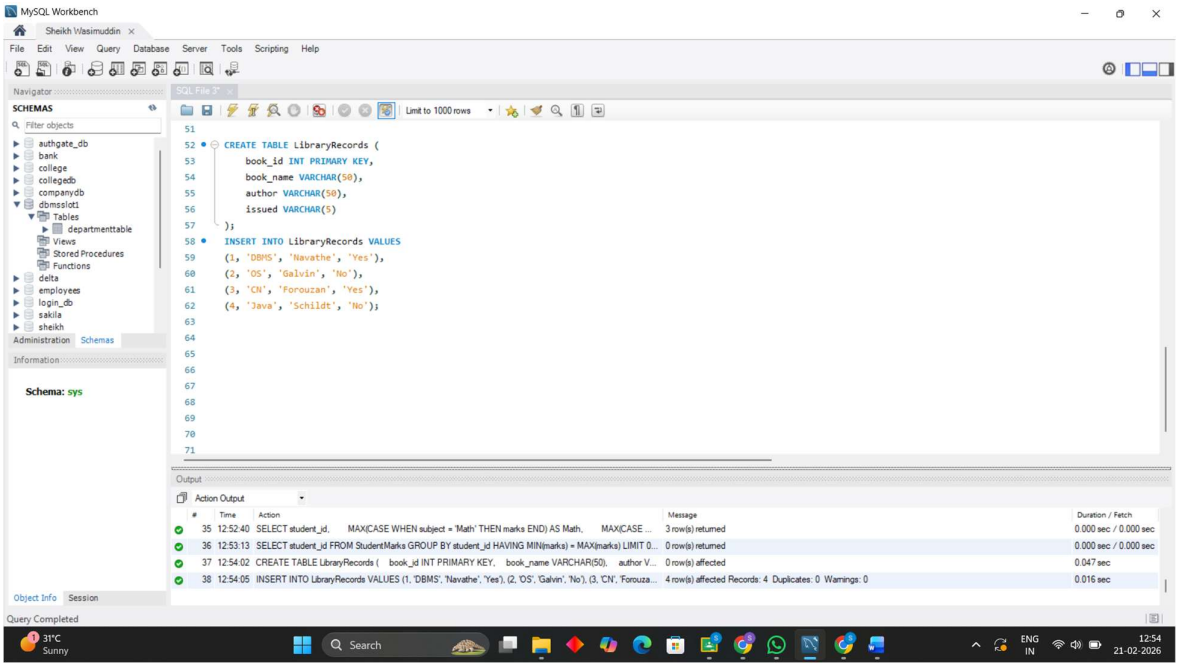
```
SELECT student_id
FROM StudentMarks
GROUP BY student_id
HAVING MIN(marks) = MAX(marks);
```

Result Grid | Filter Rows:

| student_id |
|---|

## Dataset02:LibraryRecords

| book_id | book_name | author | issued |
|---|---|---|---|
| 1 | DBMS | Navathe | Yes |
| 2 | OS | Galvin | No |
| 3 | CN | Forouzan | Yes |
| 4 | Java | Schildt | No |



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## PracticeQuestions:

### 1. Count issued and non-issued books.

```
SELECT issued, COUNT(*) AS total_books
FROM LibraryRecords
GROUP BY issued;
```

Result Grid | Filter Rows: | Export:

| issued | total_books |
|---|---|
| Yes | 2 |
| No | 2 |

### 2. Display books written by Galvin.

```
SELECT *
FROM LibraryRecords
WHERE author = 'Galvin';
```

Result Grid | Filter Rows:

| book_id | book_name | author | issued |
|---|---|---|---|
| 2 | OS | Galvin | No |
| NULL | NULL | NULL | NULL |

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## 3. Find total number of books.

```
SELECT COUNT(*) AS total_books
FROM LibraryRecords;
```

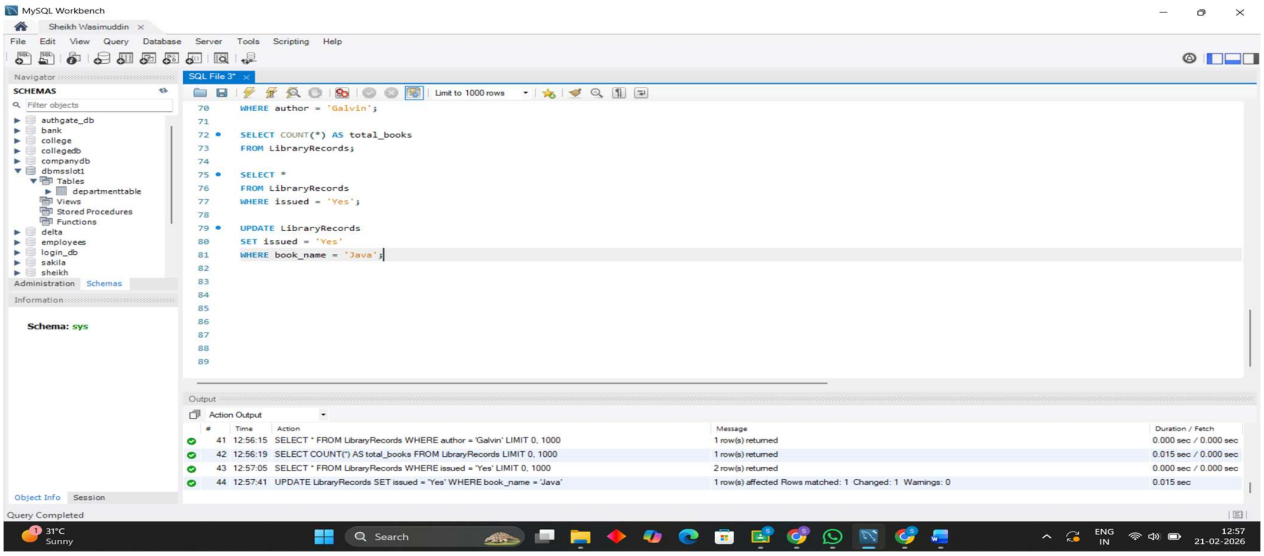
Result Grid | Filter Rows:

| total_books |
|---|
| 4 |

## 4. Show only issued books

```
SELECT *
FROM LibraryRecords
WHERE issued = 'Yes';
```

Result Grid | Filter Rows:

| book_id | book_name | author | issued |
|---|---|---|---|
| 1 | DBMS | Navathe | Yes |
| 3 | CN | Forouzan | Yes |
| NULL | NULL | NULL | NULL |

## 5. Update issued status of Java book.